

Local Energetics of Extratropical Transition: A Case Study of Typhoon Hagibis (2019)

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ABSTRACT: Tropical cyclones (TCs) moving into the midlatitudes often undergo notable structural changes as they transition into extratropical cyclones (ETCs). This transition, known as the extratropical transition (ET), results from interactions between the TC and the midlatitude baroclinic environment. To elucidate the dynamical processes underlying ET, this study proposes a cyclone-following local energetics framework and applies it to TC Hagibis (2019) in the western North Pacific. The energy cycle intensifies during the ET phase, with its spatial structure becoming markedly asymmetric relative to that in the TC phase. In both the TC and ET phases, eddy available potential energy increases as its generation and energy conversion from background flow dominate over the baroclinic energy conversion. During ET, enhanced diabatic heating, warm temperature anomalies, and vertical motion appear ahead of the cyclone due to its interaction with the midlatitude jet. This leads to an asymmetric distribution of enhanced energy sources and energy conversions for eddy available potential energy. While eddy kinetic energy also increases in the TC phase as baroclinic energy conversion exceeds lateral boundary flux, it decreases and redistributes asymmetrically during ET due to enhanced divergence of eddy geopotential flux along the boundary of the western North Pacific subtropical high and the upper-level divergent TC outflow associated with jet–TC interaction. These results show that local energetics can effectively capture asymmetric thermal and dynamical processes during ET, offering a physical interpretation of TC-to-ETC transition processes.

KEYWORDS: Extratropical transition; Tropical cyclones; Kinetic energy

1. Introduction

Tropical cyclones (TCs) that travel to the midlatitudes often undergo profound structural changes, transitioning into extratropical cyclones (ETCs). This transition, known as the extratropical transition (ET), evolves as the storm interacts with the midlatitude baroclinic environment (e.g., Jones et al. 2003; Evans et al. 2017). The ET is often associated with high-impact hazards, such as strong wind expansion (Powell 1982; Evans and Hart 2008), large ocean waves (Bowyer and MacAfee 2005), and heavy rainfall (Atallah and Bosart 2003; Chen 2011), allowing the storm to maintain or even expand its hazardous footprint well beyond the TC phase.

Previous studies have documented ET processes in various basins, including the western North Pacific (e.g., Klein et al. 2000; Kitabatake 2011), eastern Pacific (Wood and Ritchie 2014), North Atlantic (e.g., Bosart and Lackmann 1995; Hart and Evans 2001), and Southern Hemisphere (e.g., Foley and Hanstrum 1994; Sinclair 2002). Although regional differences exist, transitioning TCs exhibit a similar structural evolution. As a TC moves into the midlatitudes, it encounters decreasing sea surface temperature and increasing vertical wind shear. This leads to a weakening of its tropical characteristics, such as deep, moist, and warm-core structures. Simultaneously, the storm develops pronounced asymmetric structures, including dry intrusions, poleward-expanding cirrus shields, and frontal cloud bands (e.g., Klein et al. 2000; Harr and Elsberry 2000; Ritchie and Elsberry 2001). Consistent with these changes, the precipitation and wind fields show asymmetric evolutions (Abraham et al. 2004; Atallah et al. 2007; Fujibe and Kitabatake 2007;

Evans and Hart 2008). As the storm becomes fully embedded within the midlatitude baroclinic zone, it transitions to a distinct cold-cored extratropical system and often exhibits well-defined warm and cold sectors with pronounced frontal boundaries.

The midlatitude environment both drives and responds to the transitioning TCs (Evans et al. 2017; Keller et al. 2019). Throughout the evolution, upper-level divergent outflow of the TC is crucial, being primarily driven by diabatic heating but partially influenced by dry dynamic forcing (Grams and Archambault 2016; Quinting and Jones 2016). The divergent TC outflow modifies the background flow by advecting low potential vorticity (PV) air from the subtropics into the midlatitudes, lifting the tropopause and amplifying the downstream ridge (Bosart and Lackmann 1995; Archambault et al. 2013; Grams et al. 2013a). The enhanced PV gradient along the tropopause often anchors and intensifies the upper-level jet, while the upper-level divergent and anticyclonic PV anomaly inhibits eastward propagation of the upstream trough and wraps it up cyclonically (Agustí-Panareda et al. 2004; Riemer et al. 2008; Grams et al. 2013a; Grams and Archambault 2016). When the upstream trough and the transitioning TC are optimally phased, it can promote reintensification of a cyclone and downstream development (Ritchie and Elsberry 2003, 2007; Grams et al. 2013b; Archambault et al. 2015; Grams and Archambault 2016; Riboldi et al. 2019).

To objectively monitor the transformation of TCs into extratropical systems, the cyclone phase space (CPS) framework has been employed in both research and operations (Evans and Hart 2003; Hart 2003; Kitabatake 2011; Wood and Ritchie 2014). CPS characterizes the thermal and kinematic structure of a cyclone using parameters that represent the lower- and upper-tropospheric thermal wind and the storm asymmetry

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(Hart 2003). This framework allows the structural evolution of a TC to be interpreted in terms of its transition from a symmetric warm-core tropical system to an asymmetric cold-core extratropical system. Although CPS provides a continuous and quantitative depiction of structural evolution of a cyclone, understanding these structural changes requires a dynamical interpretation.

The present study adopts a local energetics framework to assess how TC–midlatitude interaction affects ET. Lorenz (1955) laid the groundwork for atmospheric energetics by introducing available potential energy and developing a global energy cycle. By decomposing basic variables into zonal-mean and eddy components, this framework quantifies the generation, dissipation, and conversion of kinetic energy and available potential energy. This conceptual framework, known as the Lorenz energy cycle, has become the cornerstone for understanding the fundamental energetics of the atmospheric general circulation. Building on this, local energy budget approaches further characterize the regional energy structure and evolution. Local kinetic energy budgets have been used to examine the energy dispersion in baroclinic waves, establishing the concept of downstream development (Orlanski and Katzfey 1991; Orlanski and Chang 1993; Orlanski and Sheldon 1995). Studies on locally available potential energy have linked regional energy reservoirs to storm-track activity and the thermodynamic potential for organized ascent such as warm conveyor belts (Novak and Tailleux 2018; Gertler et al. 2023).

Murakami (2011) proposed a local version of the complete Lorenz energy cycle. By introducing interaction energy and boundary flux terms, this framework distinguishes energy conversions within the local domain from energy exchanges across its boundaries, providing a local energy cycle that encompasses both the kinetic and available potential energy of the mean and eddy fields. This framework has been applied to paleoclimate (Murakami et al. 2011) and energy cascades in heavy rainfall events and their interactions with the large-scale background flow (Fu et al. 2016, 2018), demonstrating its usefulness across various temporal and spatial scales. However, Murakami's local energy cycle is formulated within an Eulerian framework, and its application to moving systems such as TCs requires reformulation into a cyclone-following framework.

In this study, local energetics during ET are examined and compared with those during the TC. To this end, the local energy cycle of Murakami (2011) is modified for use in a cyclone frame of reference. The proposed framework is applied to TC Hagibis (2019), which made landfall in Japan and caused widespread damage. TC Hagibis exhibited pronounced structural changes through interactions with the baroclinic environment in midlatitudes, making it a representative ET case (Ito and Ichikawa 2021; Yanase et al. 2022; Nayak and Takemi 2023).

The rest of this paper is organized as follows. Section 2 explains the dataset and methods, including the cyclone-following local eddy energetics. Section 3 presents an overview of TC Hagibis, including its CPS evolution and synoptic environment. Section 4 applies the local eddy energetics to ET of Hagibis. The results are summarized and discussed in section 5.

2. Data and methods

a. Dataset

The best track data of the Regional Specialized Meteorological Center Tokyo-Typhoon Center, operated by the Japan Meteorological Agency (JMA), are used for TC identification. The dataset provides at least 6-hourly (0000, 0600, 1200, and 1800 UTC) observations of longitude, latitude, central pressure, maximum sustained wind speed, storm category (i.e., tropical depression, tropical storm, typhoon, or extratropical cyclone), and wind radii for 30 and 50 kt ($1 \text{ kt} \approx 0.51 \text{ m s}^{-1}$) winds at 10 m above ground level. Satellite images from the *Geostationary Korea Multi-Purpose Satellite-2A* (GK2A) are also used to examine the structural evolution of the TC. The background fields are analyzed with the Japanese Reanalysis of Three Quarters of a Century (JRA-3Q; Kosaka et al. 2024) reanalysis data at $1.25^\circ \times 1.25^\circ$ horizontal resolution, 45 pressure levels, and 6-hourly temporal resolution. Although this spatial resolution is too coarse to resolve the detailed TC structure, it is sufficient to explore the synoptic-scale features of TC and its interaction with the large-scale environment. Although not shown, the overall results are robust to the choice of reanalysis dataset (JRA-3Q vs ERA5) and spatial resolution ($0.25^\circ \times 0.25^\circ$ vs $1.5^\circ \times 1.5^\circ$).

b. Cyclone phase space

The ET onset and completion are defined using the CPS diagram (Hart 2003; Evans and Hart 2003). The CPS employs three different parameters: the storm-relative geopotential thickness asymmetry between the right and left sides of the storm at 900–600 hPa (B), the low-level thermal wind between 900- and 600-hPa levels ($-V_T^L$), and the upper-level thermal wind between 600- and 300-hPa levels ($-V_T^U$). The thermal asymmetry of the storm B is calculated as

$$B = (\overline{Z_{600\text{hPa}} - Z_{900\text{hPa}}})_R - (\overline{Z_{600\text{hPa}} - Z_{900\text{hPa}}})_L, \quad (1)$$

where Z is the geopotential height, R and L denote the right and left sides of the storm relative to its direction of motion, and the overbar indicates the area mean within a 500-km radius of the storm center. The temperature and depth of the storm core are represented by lower-tropospheric thermal wind ($-V_T^L$), which is calculated as

$$-V_T^L = \frac{\partial \Delta Z}{\partial \ln p} \bigg|_{900\text{hPa}}^{600\text{hPa}}, \quad (2)$$

and upper-tropospheric thermal wind ($-V_T^U$), which is calculated as

$$-V_T^U = \frac{\partial \Delta Z}{\partial \ln p} \bigg|_{600\text{hPa}}^{300\text{hPa}}. \quad (3)$$

A positive $-V_T^L$ indicates a warm-core system, while a negative $-V_T^L$ indicates a cold-core system. Combined with $-V_T^U$, $-V_T^U$ offers a measure of the vertical extent of the storm core. For example, positive $-V_T^L$ and positive $-V_T^U$ indicate a deep